

PC-BASED WIRELESS CONTROL FOR TOY CAR

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Here we show how you can control a toy car through your PC's serial port using a pair of ASK transmitter and receiver modules. The received signal is decoded by a P89V51RD2 microcontroller and fed to the motor driver circuitry to move the

PC. The control signals are W, S, D, A, Q, E, C, Z and U to control the toy car in forward, backward, right drift, left drift, sharp forward left turn, sharp forward right turn, sharp backward left turn, sharp backward right turn and stop, respectively. Each of the control signals is fed from the keyboard and sent through the serial com port. The signal is then transmitted wire-

to handle.

Windows-based 9-pin serial port DB9 connector has the configuration as shown in Table I.

The RS-232 standard serial port has nine pins having different functions for transmitting and receiving data. Of these, only three pins (pins 2, 3 and

5) are mostly used for sending and receiving data. Only pins 3 and 5 are used in this project. The RS-232 standard has specific voltage levels for data logic 0 and logic 1 (-3 to -15V for logic 1, and +3 to +15V for logic 0). But the microcontroller defines logic 0 and logic

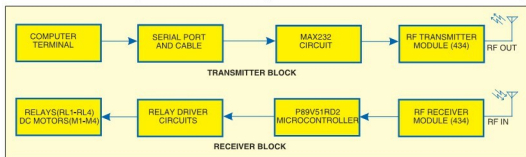


Fig. 1: Block diagram for PC-based wireless control of a toy car

toy car in forward, backward, right or left direction. All the signals are in RF domain.

Circuit description

Fig. 1 shows the block diagram for PC-based wireless control of a toy car. The different stages for wireless control are:

1. Transmission of the control signals from a PC's serial port
2. RF transmitter and receiver
3. Decoding of the received signals using the microcontroller
4. Motor drivers

Asynchronous serial communication is established between the computer and P89V51RD2 microcontroller through wireless RF link. The microcontroller and computer are both synchronised with each other. The baud rate of data transfer is 1200.

Transmission of the control signal through PC's serial port. The first part of the project is transmission of control signals through the serial port of the

TABLE I
Serial Port

Pin	Functions
1	Data-carrier detect
2	Received data
3	Transmitted data
4	Data-terminal ready
5	Signal ground
6	Data-set ready
7	Request to send
8	Clear to send
9	Ring indicator

USB port are some of the ports for connecting to the peripherals.

The serial port transmits or receives the data serially (1-bit data per TX or RX clock pulse). It is based on IEEE RS-232 standard, which defines voltages and baud rates for serial communication between devices connected to it. Most desktop computers have an RS-232 serial port as it has a very simple circuitry and is cheap and easy

lessly by the ASK transmitter module.

PC's serial com port. There are many ports available at the base of your PC in order to send data to the connected peripherals. Serial port, parallel port and

1 by voltage levels 0-0.5V and 4.5-5V, respectively. So you have to convert the RS-232 standard signal level into the microcontroller signal level. For that purpose, we have used a MAX232 converter.

For signal-level conversion, MAX232 requires four capacitors (10µF) and a 5V supply. Since we are using 9V supply, a single voltage source voltage regulator IC 7805 is required to supply power to MAX232. Now, the serial port signal, which is at RS-232 logic, can be converted into 5V TTL-level signal.

The transmitter section built around MAX232 is shown in Fig. 2.

RF transmission. The 5V supply from 7805 voltage regulator IC powers the RF ASK transmitter module through its pin 3. This module actually collects the binary signal from the data pin (which is connected to the MAX232) and modulates this binary signal with amplitude-shift keying (ASK) digital modulation scheme by

